

Business Economics

Marcelo Ortiz M.

Topic 6: Performance Evaluation

Objectives

- 1) understand the different dimensions of performance
- 2) identify appropriate evaluation methods
- 3) list examples of how an employee can game performance measurement
- 4) explain the benefits and cost of relative performance evaluation
- 5) explain the benefits and cost of and subjective performance evaluation

Why performance evaluation is needed?

1) to provide feedback on job achievements that allows improvements

Additional training, failures points

2) to determine rewards and sanctions

Wage, rises, bonuses, promotion, dismissals

Notice that both roles create different incentives for employees.

1) no incentive to distort the evaluation

2) incentive to distort the evaluation

In this topic, we will focus on its role in rewarding

Principal-Agent Model: revisited

Employee's output "Q" depends on effort "e" and a random component "μ":

$$Q = \alpha e + \mu$$

Where "α" is the employee's marginal productivity known by the management and the employee, and "e" and "μ" are not observable by management

If the employee received a fixed compensation scheme independent of the output, he has incentives to shirk because low Q can be blamed on negative μ

To limit this behavior, the compensation contract is based on the output

$$\text{Employee Compensation} = W_0 + \delta Q$$

Where δ is the sensitivity of pay to performance.

Principal-Agent Model: revisited

This compensation contract reduces shirking on effort but also impose risk on employee because its payment is a function of μ

Since the employee is risk-averse, the firm must compensate him for bearing the risk, or the employee will choose to work elsewhere

Difficulties on measuring performance Q

Performance (or Output) Q can be:

- **Observable (fruit pickers) or non-observable at the moment of evaluation.**
 - **If observable, then is contractible.**
- **Multidimensional**

Types of non-observable:

- **because its intangible nature (Brand Specialist, R&D Researcher)**
- **because require time to materialize (Product Design)**
- **because Q depends on a team or dependencies across employees.**

Measurement cost

When employee's output is not observable in a reasonable time period, the performance evaluation becomes more complicated.

Patent attorneys

School teachers

Lab researcher

Performance evaluation can be quite costly:

Technology: workers' effort and heart monitors

Specific-knowledge: hiring top leading scientists to evaluate the performance of lab researchers

Measurement cost

But which is the benefit of using costly measures for evaluating performance?

It reduces the compensation differential the firm must pay the employee to bear the risk of being evaluated with exogenous random risk.

Therefore, a value-maximizing firm will go to the point where the incremental cost of an additional performance measure equals the incremental benefits (lower risk-premium compensation)

If the Q measure is really too costly, firms rely on proxies (Y).

However, this might generate opportunism: short-termism and gaming

Measurement cost

Short-termism: enhancing current-period Y at the cost of reducing the long-term Q

Ex: a CEO cutting R&D expenses for beating current profit benchmark

Gaming: if Q is not measured and the benchmark is another variable Y, then the incentive is to increase Y (and not necessarily Q)

Ex: policeman arresting drivers unnecessarily because his evaluation considers the number of arrests (Y) while the real outcome is the number of reckless drivers arrested (Q). Q and Y are correlated, but are not the same

Measuring Q: Setting Performance Benchmark

(a) Time and motion studies: systematic analysis of work methods, ranging from the raw material and each step until deliver of the final product.

How long does each activity take?

Work sampling

Example: process of delivering a passport in a civil registry. Assembly lines.

These types of studies are expensive

Engineers time, need to be updated frequently

Incentives? Bias from employees to generate worse metric during the study

Measuring Q: Setting Performance Benchmark

(b) Past performance and the Ratchet effect

Another way to define a benchmark is by using past years performance

But this might not be a clever idea: the Ratchet effect refers to basing next year's benchmark on this year's performance.

The problem: benchmark is more likely to move up or not move at all (not adjusted by bad year, not symmetric).

This effect discourages employees from exceeding the benchmark substantially to avoid raising the standard for future years

Example: discuss the incentives created by the following benchmarks if are used in compensation contracts:

- 1) last year's sales for a salesperson's bonus
- 2) an assembly plant punishing employees not meeting past's year production but rewarding those who beat it

Measuring Q: Setting Performance Benchmark

Potential solutions:

- 1) Once the benchmark is defined, it is not changed until a new production/selling technology is implemented, no matter how much money the employees obtain.
- 2) Rotation

Measuring Q: Relative Performance Evaluation

a) Within-firm performance

Employees within the same unit/firm face the same risk factors: economic fluctuations, competitors, quality of inputs, etc

Therefore, using peers from within the unit as a benchmark for evaluating performance allows stripping the common risk factor from the observed performance

If a salesperson typically sells \$1,000 in a month, but their sales for the month in question are \$700, the sales manager can evaluate other salespersons' performances to see if they also experienced a 30% reduction in sales.

Problem 1: punishment of rate busters, sabotage, and collusion

Problem 2: incentives for recruiting worse candidates (to improve the relative performance of current employees)

Measuring Q: Relative Performance Evaluation

b) Across-firm performance

It's useful when there is a lack of an internal reference group (CEOs) or the incentives for pernicious actions of sabotage and collusion are high

The information can be shared directly or indirectly (trade associations and consultants)

Problems:

- Antitrust law and competition

- Firms might have different assignments of decision rights for the same position

- External shocks in employees' performance can be unrelated

Measuring Q: Subjective Performance Evaluation

Many important dimensions of performance can not be objectively measured

Ex: mentoring and team collaboration

If third parties cannot verify measures of these dimensions, they are considered subjective measures

Subjective measures allow a more holistic evaluation of the employee's performance

A complete objective performance evaluation is too costly

In practice, most employees receive a subjective performance evaluation from their supervisors in annual meetings/review

Measuring Q: Subjective Performance Evaluation

Methods:

a) standard-rate scale system

The evaluator ranks (0-5) the employee using performance factors such as organizational efficiency, teamwork, achievements, identification and solution of new problems.

Then the evaluator assigns an overall rank and provides suggestion

b) goal-based system

The goals must inform the evaluator about subjective measures of performance (ex, teamwork)

The goals tend to be easy to measure

Ex: "No complain from group peers until 1 December", "create and hold a monthly mentoring session for junior colleagues"

The evaluator considers the degree of general achievement of the set of goals

Measuring Q: Subjective Performance Evaluation

Problems:

Different evaluators, different criteria

Employee responses or explanation might be foregone

Training supervisors to become evaluators

Collusion and friendships

Shirking among supervisors and inaccurate measures

Influence cost (again!)

Measuring Q:

Combination of performance measure

Remember the informativeness principle

Although both objective and subjective performance measures can be inaccurate measures of the contribution to the firm value of each employee, they increase the accuracy in their performance evaluation

More accuracy → less risk-bearing for the employee → lower the risk compensation the firm must pay

However, as discussed before, they can also lead to dysfunctional behaviors

The ideal compensation contract includes a set of performance metrics that reduce the likelihood or the cost of such undesirable behaviors.

Summary

Main concepts:

Need for evaluating performance

Time and motion studies

Ratchet effect

Measurement cost

Short-termism and Gaming

Standard-rate scale system and goal-based system

Cases to Discuss

Present an real case related to some of the topics of the class.

The case should be based on recent news (<2 years) and preferably related to Spanish companies/cases.

Start the presentation with the context behind the event/decision/case, then identify the relevant concepts, to finally conclude/assess the case.

10 min, 7-11 slides

Examples:

- 1) Short-termism and firm cutting investment/stopping projects
- 2) Gaming and accounting manipulation/scandals in Spain/EU
- 3) Combined methods: Cases of 360-performance evaluation in Spain/EU